phosphorus in the plant; it was found to be about 10 cm. per second.

Artificially produced radioactive substances, and, again, radioactive phosphorus in particular, have been used to study animal metabolism. In these cases, the phosphorus was mixed with the food of the animals or injected into their bodies (*Hevesy*). This technique makes it possible to determine, after a while, in what parts of the organism the phosphorus has a tendency to settle, and also the speed of its elimination, so that information concerning metabolism can be obtained not only numerically and as to proportion, but also as to details. Among other things, it has been ascertained that after a certain length of time, the phosphorus settles principally in the bones and in the liver, and after another period of time, in the teeth. A great many important biological data can be obtained in this manner.

The quantities of radioactive substance required for such experiments are so minute that they cannot result in any harm to the organism.

Another important problem which could be tackled by the tracer method was that of the assimilation by plants of carbon dioxide. It is a well-known fact that green plants utilize the effect of sunlight (in other words, a photochemical process) to assimilate carbon dioxide from the air and convert it into hydrocarbons. This is how they store up solar energy. But very little was known about the details of this process, and various theories were formulated about its mechanism. It was known for a certainty that every plant must absorb approximately four photons of light so as eventually, with the aid of their energy, to accomplish the chemical reaction. In order to explain this problem, the American scientists Ruben, Hassid and Kamen used carbon dioxide containing radioactive carbon of mass number 11, which has a half-life of 20 minutes. They found that in order to prepare the way for an assimilation, first a reaction in darkness takes place, in the course of which both the carbon and the oxygen of the carbon dioxide (CO_2) are combined with the addition of hydrogen in the form of the carboxyl group (of the residue COOH) in a giant organic molecule. Later on, glucose (grape sugar, $C_6H_{12}O_6$) is formed from the carboxyl group, possibly by the roundabout way of larger sugar molecules. As a result, the important fact was established that the process of assimilation takes place in several stages. However, we shall not enlarge on further details of this subject here.

VIII. ARTIFICIAL RADIOACTIVE SUBSTANCES IN MEDICINE

Artificial radioactive substances are also destined to render valuable service at a later date in medicine. In contrast to the experiments described above, where the important thing is the investigation of the normal unimpaired organism, by the aid of small quantities of radioactive materials, it is also